

Commerce XI Maths App

Unit 3 Theory (Pages 33-37)

Detailed answers with explanations and steps

Total questions: 38 | +2 correct | -1 wrong

Question 1 - Theory P33

The symbol i is introduced so that

Options

A. $i = \sqrt{-1}$ and $i^2 = -1$

B. $i = -1$

C. $i^2 = 1$

D. $i = \sqrt{1}$

Correct answer: $i = \sqrt{-1}$ and $i^2 = -1$

Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives $i = \sqrt{-1}$ and $i^2 = -1$.

Question 2 - Theory P33

The symbol i is called

Options

A. an imaginary unit

B. a real unit

C. a rational unit

D. a natural unit

Correct answer: an imaginary unit

Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives an imaginary unit.

Question 3 - Theory P33

A number of the form ki , where $k \in \mathbb{R}$, $k \neq 0$ and $i = \sqrt{-1}$, is called

Options

A. an imaginary number

B. a natural number

C. a rational number

D. a real number

Correct answer: an imaginary number

Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives an imaginary number.

Question 4 - Theory P33

Write $\sqrt{-25}$ using i

Options

- A. $5i$
- B. $-5i$
- C. $25i$
- D. 5

Correct answer: $5i$

Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives $5i$.

Question 5 - Let's Note P33

For $a \in \mathbb{R}$, $\sqrt{-a^2}$ equals

Options

- A. $\pm ia$
- B. a
- C. $-a$
- D. i^2a

Correct answer: $\pm ia$

Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives $\pm ia$.

Question 6 - Let's Note P33

If $a, b \in \mathbb{R}$ and $ai = bi$, then

Options

- A. $a = b$
- B. $a = -b$
- C. $a + b = 0$ always
- D. $ab = 1$

Correct answer: $a = b$

Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives $a = b$.

Question 7 - Theory P33

A number of the form $a + ib$, where $a, b \in \mathbb{R}$, is called

Options

- A. a complex number
- B. an irrational number
- C. a natural number
- D. a whole number

Correct answer: a complex number

Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives a complex number.

Question 8 - Theory P33

The set of complex numbers is denoted by

Options

- A. $\mathbb{C}$
- B. $\mathbb{R}$
- C. $\mathbb{I}$
- D. $\mathbb{N}$

Correct answer: C

Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives C.

Question 9 - Theory P33

For $z = a + ib$, $\operatorname{Re}(z)$ equals

Options

- A. a
- B. b
- C. ib
- D. $a + b$

Correct answer: a

Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives a.

Question 10 - Theory P33

For $z = a + ib$, $\text{Im}(z)$ equals

Options

- A. b
- B. a
- C. ib
- D. $a + b$

Correct answer: b

Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives b .

Question 11 - Example P33

For $z = 2 + 4i$, select $\text{Re}(z)$ and $\text{Im}(z)$

Options

- A. $\text{Re}(z) = 2, \text{Im}(z) = 4$
- B. $\text{Re}(z) = 4, \text{Im}(z) = 2$
- C. $\text{Re}(z) = 2, \text{Im}(z) = 4i$
- D. $\text{Re}(z) = 0, \text{Im}(z) = 4$

Correct answer: $\text{Re}(z) = 2, \text{Im}(z) = 4$

Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives $\text{Re}(z) = 2, \text{Im}(z) = 4$.

Question 12 - Example P33

Write $5 + \sqrt{-16}$ in the form $a + ib$

Options

- A. $5 + 4i$
- B. $5 - 4i$
- C. $9i$
- D. $5 + 16i$

Correct answer: $5 + 4i$

Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives $5 + 4i$.

Question 13 - Example P33

For $z = 7 + \sqrt{3}$, select $\operatorname{Re}(z)$ and $\operatorname{Im}(z)$

Options

- A. $\operatorname{Re}(z) = 7 + \sqrt{3}$, $\operatorname{Im}(z) = 0$
- B. $\operatorname{Re}(z) = 7$, $\operatorname{Im}(z) = \sqrt{3}$
- C. $\operatorname{Re}(z) = 7 + \sqrt{3}$, $\operatorname{Im}(z) = 1$
- D. $\operatorname{Re}(z) = 0$, $\operatorname{Im}(z) = 7 + \sqrt{3}$

Correct answer: $\operatorname{Re}(z) = 7 + \sqrt{3}$, $\operatorname{Im}(z) = 0$

Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives $\operatorname{Re}(z) = 7 + \sqrt{3}$, $\operatorname{Im}(z) = 0$.

Question 14 - Let's Note P34

A complex number whose real part is zero is

Options

- A. an imaginary number
- B. a real number
- C. always zero
- D. a natural number

Correct answer: an imaginary number

Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives an imaginary number.

Question 15 - Let's Note P34

A complex number whose imaginary part is zero is

Options

- A. a real number
- B. an imaginary number
- C. not a complex number
- D. always non-real

Correct answer: a real number

Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives a real number.

Question 16 - Let's Note P34

The zero complex number is written as

Options

- A. $0 + 0i$
- B. $1 + 0i$
- C. $0 + i$
- D. $i + i$

Correct answer: $0 + 0i$

Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives $0 + 0i$.

Question 17 - Let's Note P34

Which inclusion is correct?

Options

- A. $\mathbb{R} \subset \mathbb{C}$
- B. $\mathbb{C} \subset \mathbb{R}$
- C. $\mathbb{R} = \mathbb{C}$
- D. $\mathbb{R} \cap \mathbb{C} = \emptyset$

Correct answer: $\mathbb{R} \subset \mathbb{C}$

Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives $\mathbb{R} \subset \mathbb{C}$.

Question 18 - Theory P34

The conjugate of $z = a + ib$ is

Options

- A. $\bar{z} = a - ib$
- B. $\bar{z} = -a + ib$
- C. $\bar{z} = b + ia$
- D. $\bar{z} = a + ib$

Correct answer: $\bar{z} = a - ib$

Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives $\bar{z} = a - ib$.

Question 19 - Example P34

Find the conjugate of $3 + 4i$

Options

- A. $3 - 4i$
- B. $-3 + 4i$
- C. $-3 - 4i$
- D. $4 - 3i$

Correct answer: $3 - 4i$

Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives $3 - 4i$.

Question 20 - Example P34

Find the conjugate of $7i - 2$

Options

- A. $-2 - 7i$
- B. $2 - 7i$
- C. $-2 + 7i$
- D. $7 - 2i$

Correct answer: $-2 - 7i$

Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives $-2 - 7i$.

Question 21 - Theory P34

For every complex number z , which statement is true?

Options

- A. $\bar{z} = z$
- B. $\bar{z} = z$ always
- C. $\bar{z} = -z$ always
- D. $z + \bar{z} = 0$ always

Correct answer: $\bar{z} = z$

Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives $\bar{z} = z$.

Question 22 - Theory P34

If $z = \bar{z}$, then z is

Options

- A. real
- B. imaginary
- C. zero only
- D. non-real

Correct answer: real

Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives real.

Question 23 - Theory P34

If $z = -\bar{z}$, then z is

Options

- A. imaginary
- B. real
- C. positive
- D. natural

Correct answer: imaginary

Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives imaginary.

Question 24 - Theory P34

If $a + ib = c + id$, then

Options

- A. $a = c$ and $b = d$
- B. $a = d$ and $b = c$
- C. $a + b = c + d$ only
- D. $a = -c$ and $b = -d$

Correct answer: $a = c$ and $b = d$

Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives $a = c$ and $b = d$.

Question 25 - Example P34

If $x + iy = 4 + 3i$, then

Options

- A. $x = 4$ and $y = 3$
- B. $x = 3$ and $y = 4$
- C. $x = 4$ and $y = 3i$
- D. $x = -4$ and $y = -3$

Correct answer: $x = 4$ and $y = 3$

Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives $x = 4$ and $y = 3$.

Question 26 - Theory P34

If $z_1 = a + ib$ and $z_2 = c + id$, then $z_1 + z_2$ equals

Options

- A. $(a + c) + (b + d)i$
- B. $(a - c) + (b - d)i$
- C. $(ac - bd) + (ad + bc)i$
- D. $(a + b) + (c + d)i$

Correct answer: $(a + c) + (b + d)i$

Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives $(a + c) + (b + d)i$.

Question 27 - Theory P34

For complex numbers z_1 and z_2 , $\operatorname{Re}(z_1 + z_2)$ equals

Options

- A. $\operatorname{Re}(z_1) + \operatorname{Re}(z_2)$
- B. $\operatorname{Re}(z_1)\operatorname{Re}(z_2)$
- C. $\operatorname{Im}(z_1) + \operatorname{Im}(z_2)$
- D. $\operatorname{Re}(z_1) - \operatorname{Re}(z_2)$

Correct answer: $\operatorname{Re}(z_1) + \operatorname{Re}(z_2)$

Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives $\operatorname{Re}(z_1) + \operatorname{Re}(z_2)$.

Question 28 - Theory P34

For a complex number z , $z + \bar{z}$ equals

Options

- A. $2\operatorname{Re}(z)$
- B. $2\operatorname{Im}(z)$
- C. 0 always
- D. z^2

Correct answer: $2\operatorname{Re}(z)$

Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives $2\operatorname{Re}(z)$.

Question 29 - Theory P34

For a real scalar k and $z = a + ib$, kz equals

Options

- A. $ka + i(kb)$
- B. $ka + ib$
- C. $a + ik$
- D. $kab + i$

Correct answer: $ka + i(kb)$

Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives $ka + i(kb)$.

Question 30 - Theory P34-35

If $z_1 = a + ib$ and $z_2 = c + id$, then $z_1 - z_2$ equals

Options

- A. $(a - c) + (b - d)i$
- B. $(a + c) + (b + d)i$
- C. $(a - c) - (b - d)i$
- D. $(ac - bd) + (ad + bc)i$

Correct answer: $(a - c) + (b - d)i$

Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives $(a - c) + (b - d)i$.

Question 31 - Example P35

If $z_1 = 4 + 3i$ and $z_2 = 2 + i$, find $z_1 - z_2$

Options

- A. $2 + 2i$
- B. $2 + 4i$
- C. $6 + 4i$
- D. $-2 - 2i$

Correct answer: $2 + 2i$

Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives $2 + 2i$.

Question 32 - Theory P35

If $z_1 = a + ib$ and $z_2 = c + id$, then $z_1 z_2$ equals

Options

- A. $(ac - bd) + (ad + bc)i$
- B. $(ac + bd) + (ad - bc)i$
- C. $(a + c) + (b + d)i$
- D. $(ac - bd) + (ab + cd)i$

Correct answer: $(ac - bd) + (ad + bc)i$

Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives $(ac - bd) + (ad + bc)i$.

Question 33 - Example P35

Find $(2 + 3i)(3 - 2i)$

Options

- A. $12 + 5i$
- B. $12 - 5i$
- C. $5 + 12i$
- D. $6 + 5i$

Correct answer: $12 + 5i$

Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives $12 + 5i$.

Question 34 - Let's Note P35

If $z = a + ib$, then $z \cdot \bar{z}$ equals

Options

A. $a^2 + b^2$

B. $a^2 - b^2$

C. $2ab$

D. $a + b$

Correct answer: $a^2 + b^2$

Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives $a^2 + b^2$.

Question 35 - Let's Note P35

The values of i^2 , i^3 and i^4 are respectively

Options

A. $-1, -i, 1$

B. $1, i, -1$

C. $-1, i, 1$

D. $i, -1, -i$

Correct answer: $-1, -i, 1$

Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives $-1, -i, 1$.

Question 36 - Let's Note P35

For $n \in \mathbb{N}$, i^{4n+2} equals

Options

A. -1

B. 1

C. i

D. $-i$

Correct answer: -1

Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives -1 .

Question 37 - Theory P35

Division by $c + id$ is carried out by multiplying numerator and denominator by

Options

- A. $c - id$
- B. $c + id$
- C. $-c + id$
- D. $i(c + d)$

Correct answer: $c - id$

Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives $c - id$.

Question 38 - Solved Example P36

If real numbers a and b satisfy $(i^4 + 3i)a + (i - 1)b + 5i^3 = 0$, then

Options

- A. $a = b = 5/4$
- B. $a = b = 4/5$
- C. $a = 5/4, b = -5/4$
- D. $a = 0, b = 5$

Correct answer: $a = b = 5/4$

Explanation and steps

This follows from the definition, identity, or solved example stated in Unit 3. Applying that result gives $a = b = 5/4$.